

Classifying Parliamentary Text into Ministries

A Study of Representation Methods

DS5601: Natural Language Processing

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Problem Motivation

- Parliamentary proceedings are released as long, unstructured documents.
- Important information (topics, ministries) is not explicitly labeled.
- **Goal:** Identify which ministry a paragraph belongs to.
- **Key Challenge:**
 - No annotated dataset
 - High domain complexity
- **Core Idea:**
 - Explore different paradigms of text representation
 - Evaluate methods without relying on gold labels

Data and Setup

- **Data:**

- Parliamentary transcripts (debates, budget speeches, Q&A)
- 53 ministry knowledge bases (Wikipedia + Govt + LLM summaries)

- **Processing:**

- Speaker-wise and paragraph-level segmentation
- Cleaning + domain-specific stopwords

- **Evaluation:**

- No ground truth labels
- Weak supervision using LLM + cross-method comparison

Topic Modeling (Supporting Structure)

- Topic modeling performed using **David Mimno's online tool**
<https://mimno.infosci.cornell.edu/jsLDA/>
- **Outputs obtained:**
 - 25 topics over the full corpus
 - Top 10 representative words per topic
 - Document–Word–Topic state distributions
 - Automatically generated domain-specific stopwords

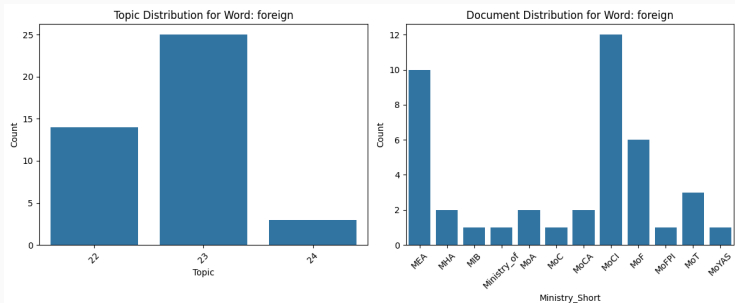


Figure 1: Example of word–topic and word–document relationships

Three Ways to Represent Text

Method	Core Idea
Lexical (TF-IDF)	Meaning = Words
Probabilistic	Words $\rightarrow$ Topics $\rightarrow$ Ministries
Embedding-based	Meaning = Vector similarity

Research Question: Which representation works best when no labels are available?

Method 1: Lexical (TF-IDF)

Idea: Match paragraph words with ministry-specific vocabulary

Assumption:

- Words directly indicate meaning

Works well when:

- Distinct domain keywords exist (e.g., “fertilizer”, “crop”)

Fails when:

- Synonyms are used (“budget” vs “fiscal policy”)
- Vocabulary overlaps across ministries

Insight: Parliamentary language is not discriminative enough for pure keyword methods.

Method 2: Probabilistic (Doc-Word-Topic)

Idea: Words $\rightarrow$ Topics $\rightarrow$ Ministries

$$\text{Score}(d) = \sum_w \sum_t P(d|w) \cdot P(t|w) \cdot P(d|t)$$

Assumptions:

- Words are independent
- Topics explain word usage
- Frequencies approximate probabilities

Strength: Combines corpus statistics with structure

Fails because:

- Independence assumption is violated
- Topics overlap across ministries

Insight: Real language does not follow clean probabilistic assumptions.

Method 3: Embedding-Based

Idea: Represent text as vectors and compare using similarity

Variants:

- Whole-document embedding (loses detail)
- Mean of paragraph embeddings (**best**)
- Top-K similar paragraphs (captures local similarity)

Why it works:

- Captures semantic meaning beyond keywords
- Handles synonyms and paraphrases

Limitations:

- Sensitive to representation design
- Can blur multiple topics

Embedding Space Visualization



Figure 2: t-SNE projection

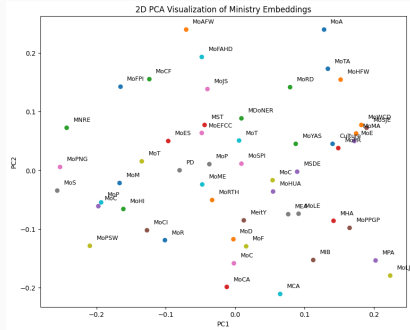


Figure 3: PCA projection

Observations:

- Some ministries form clear clusters
- Significant overlap exists between related ministries

Clustering Ministries

- Observed weak separation (low silhouette scores) for hard clustering
- Adopted **consensus-based clustering**: Group ministries based on stable co-occurrence patterns

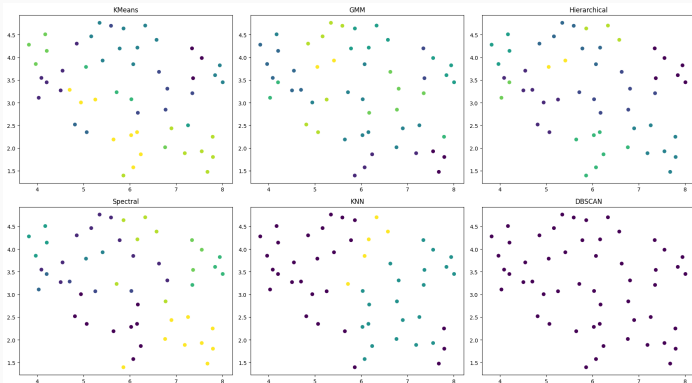


Figure 4: Consensus-based grouping of ministries (Plotted using UMAP)

Comparison with Weak Supervision (ChatGPT Labels)

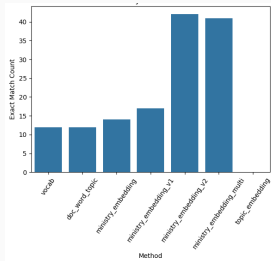


Figure 5: Exact Match

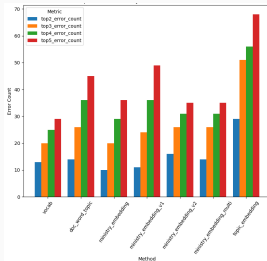


Figure 6: Top-K Errors

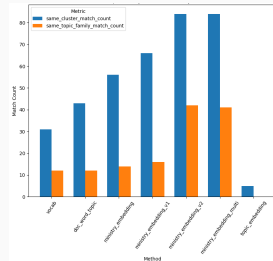


Figure 7: Cluster Match

Observations:

- Embedding methods show higher consistency across metrics
- Probabilistic method improves over lexical baseline
- Exact match is low, but cluster-level agreement is higher

Insight: In absence of ground truth, consistency across views is more meaningful than raw accuracy.

Results and Key Observations

- Embedding-based methods outperform lexical approaches
- Probabilistic method improves over simple keyword matching
- Best performance: **paragraph-level embedding averaging**

Important Observation:

- No method is perfectly correct
- Some methods are more **consistent** than others

Reason:

- High semantic overlap between ministries

Conclusion and Takeaways

- Ministry classification is fundamentally a **representation problem**
- **Lexical methods:**
 - Fail due to vocabulary overlap and lack of context
- **Probabilistic methods:**
 - Limited by unrealistic independence and topic assumptions
- **Embedding methods:**
 - Capture semantics effectively
 - Performance depends on representation design (granularity)

Key Insight: Better models did not solve the problem
— better representations did.

Can You Predict the Ministry?

Sample processed paragraph text:

prime dhandhaanya krishi yojana developing agri districts programme
partnership states schemes measures cover low productivity moderate
crop intensity credit enhance agricultural diversification sustainable
agriculture harvest storage panchayat irrigation

What our methods predicted—

Method	Predicted Ministry	Score
TF-IDF (Lexical)	Agriculture and Farmers Welfare	0.0680
Probabilistic	Agriculture and Farmers Welfare	0.2049
Embedding (Full)	Cooperation	0.0228
Embedding (Mean)	Agriculture and Farmers Welfare	0.0249
Embedding (Top-K)	Agriculture and Farmers Welfare	0.0238
Topic Embeddings	Consumer Affairs (F&PD)	0.0856
ChatGPT	Agriculture and Farmers Welfare	—

Can You Predict the Ministry?

Sample processed paragraph text:

post catalyst rural economy lakh offices complemented payment bank
vast network dak sevaks repositioned

What our methods predicted—

Method	Predicted Ministry	Score
TF-IDF (Lexical)	Communication	0.1212
Probabilistic	Communication	0.4200
Embedding (Full)	Housing and Urban Affairs	0.0223
Embedding (Mean)	Rural Development	0.0229
Embedding (Top-K)	Communications	0.0219
Topic Embeddings	Consumer Affairs (F&PD)	0.0811
ChatGPT	Communications	—

Observation: Even simple examples can lead to different predictions across methods.

Thank You

Questions?

Project Repository:

https://github.com/DevadattaP/analyse_parliament_transcripts_NLP